

Signed Quadratic-Feedback Inversion

A finite-core resolution and sharp entire Borel growth

Research article prepared for Vladimir Reshetnikov
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Abstract

The ProveIt article on finite-core universality leaves conjectural a signed coefficient equivalent for the compositional inverse of $\hat{U}(q) = \sum_{j \geq 1} q^j \exp(aj^2 \hat{U}(q))$, $a > 0$. We give a proof by reversing the defining equation before performing large-order analysis. The crucial estimate is an arbitrary-algebraic-accuracy reduction to one action above a sufficiently large analytic core. Its proof uses absolute bounds only after the cancellations within the core have been resummed. A merger defect for quadratic actions suppresses all remaining multiple-tail terms.

If $r_n(1+r_n)e^{2r_n} = an$ and $S_n = \exp(nr_n(1+2r_n)/(1+r_n))/\sqrt{2r_n^2 + 4r_n + 1}$, the result is $v_n \sim -S_n \exp(-(1+1/a)r_n)$. More generally, arbitrary real changes to finitely many weights and slopes preserve the equivalent, with $1+1/a$ replaced by $(\lambda_1 + w_2)/a$. Thus eventual negativity survives finite signed perturbations. Although a large auxiliary core is used in the proof, the leading inverse equivalent needs only the first two actions; the one-action core is insufficient in the pure model.

A second saddle calculation determines the entire ordinary Borel transform: if $s(s+1) = aR$ and $b = (\lambda_1 + w_2)/a$, then $-\mathcal{B}\hat{V}(R) \sim \exp(se^{2s}/a - bs)/\sqrt{2s+1}$. This is also its maximum-modulus equivalent. It locates the positive-direction Borel obstruction at infinity, not at a finite Borel singularity. The article includes complete estimates, exact finite identities, reproducible checks, scope limitations, and twelve further research questions.

Proof and novelty status. This is a conventional mathematical research draft, not a Lean-verified or independently refereed paper. The principal result resolves a precisely identified *repository-local* conjecture. No claim of worldwide publication priority or of solving unrestricted transseries summability is made. The central estimates are proved here rather than imported from the repository's positive-coefficient arguments.

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1 The question and the change of viewpoint

1.1 The exact repository target

The ProveIt transseries collection develops formal inversion, support control, large-order estimates, analytic realization, and quantitative asymptotics. Its manuscript *Finite-Core Universality and Sharp Large Order for Countable Exponential-Feedback Transseries* explicitly labels

$$v_n \sim -S_n \exp(-(1 + 1/a)r_n) \quad (1.1)$$

as the “Quadratic inverse equivalent” conjecture (TeX label `conj:quadratic-inverse`) [1]. The manuscript proves the corresponding sharp inverse statement for superquadratic feedback, but explains why that proof stops at the quadratic boundary: its expansion parameter grows rather than tends to zero. Its exact one-tail inverse-response identity suggests (1.1), while leaving dominance of that response unproved.

We address that missing dominance step. The point is not to compute more terms in the inverse, nor to extrapolate a positive forward-coefficient argument across a sign change. Instead, put $t = \widehat{U}(q)$ and invert the *defining equation*:

$$t = \sum_{j \geq 1} w_j q^j e^{\lambda_j t}, \quad q = \widehat{V}(t). \quad (1.2)$$

At fixed t , the feedback exponentials are now coefficients of a power series in q . A finite part of that series can be inverted analytically. Only the remaining large actions are expanded. This separates the cancellation-producing background from the combinatorial tail estimate.

The inspected repository commit was `0c973d8f5e7ef4550f4df1bf486d890037fd9f14`. The relevant `article.tex` blob was `c2b6faf623003111e9d9e6bbb162a67429f3ce42`. The bibliography and package provenance file give the exact path. The inspection was targeted, not an exhaustive audit of all branches or incoming research archives.

Editorial note (ProveIt, 2026-09-29). The claimed proof below has not been independently reviewed; a second, independent claimed proof of the same conjecture is [6] (see the note after Theorem 2.2). The inspected snapshot also contained two articles that this one does not cite. The negative-ray article [7] asks, in its research question “Signed coefficient asymptotics and actual least error” (`Negative_Ray_Summation_Exponential_Feedback/article.tex`, lines 1372–1381), for a signed asymptotic equivalent of the inverse coefficients in the quadratic case; Theorem 2.2 answers that part for the inverse (the case $a = 1$ with no exceptions), while the series $\widehat{P}$ and the actual least truncation error there are not treated. The regularity article [8], Theorem `thm:borelintro` (`Exponential_Feedback_Regularity_Classification/article.tex`, lines 288–303), proves the forward analogue of Theorem 2.4, $\log \log B_{\mathcal{U}}(t) \sim 2\sqrt{t}$ for $\lambda_j = j^2$, the same rate as (2.14) at $a = 1$. The later weighted-type package [9] (batch 48), which could not be seen from this snapshot, proves for $\lambda_j \sim aj^2$ that the normalized roots $(|v_n| \log(n+e)^{2n}/n!)^{1/n}$ have $\limsup 4a$ and that $\limsup \log \log \max_{|z|=R} |B(z)|/\sqrt{R} = 2\sqrt{a}$ (4 and 2 for $\lambda_j = j^2$); in the pure model Corollary 7.1 and (2.14) upgrade both limsups to limits with the same values.

1.2 Established tools and the proposed contribution

Lagrange inversion is classical; we use its formal and parameter-dependent versions in the form surveyed by Gessel [2]. Transseries supply the formal asymptotic setting, not an automatic assertion of analytic convergence [5]. The distinction between formal Borel transforms, analytic continuation, and Laplace growth is standard in summability theory [4].

Borinsky’s algebra of factorially divergent series provides an important comparison [3]. Our coefficients are smaller than every fixed positive exponential multiple of $n!$, even after multiplying

by any power of n . Consequently they have the *zero* asymptotic expansion in every fixed positive- α gamma scale used there. They are not excluded from those classes; rather, those particular asymptotic invariants do not detect the moving logarithmic scale resolved here.

The proposed contributions are the signed quadratic equivalent, its finite-perturbation universality, the sharp two-action leading core, and the explicit maximum-modulus equivalent for the inverse Borel transform. The underlying inversion formula, Gaussian integral, and Cauchy estimates are not claimed as new.

2 Model, notation, and main theorems

Definition 2.1 (Eventually quadratic feedback). Fix $a > 0$. Let $w_j, \lambda_j \in \mathbb{R}$, $j \geq 1$, satisfy $w_1 = 1$ and

$$w_j = 1, \quad \lambda_j = aj^2 \quad (j > M_0) \quad (2.1)$$

for some fixed integer M_0 . No positivity assumption is imposed on the finitely many exceptional weights or slopes. Define $\hat{U} \in q\mathbb{R}[[q]]$ by

$$\hat{U}(q) = \sum_{j \geq 1} w_j q^j e^{\lambda_j \hat{U}(q)}, \quad \hat{V} = \hat{U}^{(-1)} = \sum_{n \geq 1} v_n t^n. \quad (2.2)$$

Set

$$c = \lambda_1 + w_2, \quad b = c/a, \quad C_0 = 1 + |\lambda_1| + |w_2|. \quad (2.3)$$

All parameters and all core sizes in this article are fixed while n tends to infinity.

The right side of (2.2) is locally finite in the q -adic filtration. Its coefficient of q^n depends only on coefficients of $\hat{U}$ of degrees less than n , because every summand has a factor q^j . This proves existence and uniqueness recursively. Also $\hat{U}(q) = q + cq^2 + O(q^3)$, so its formal compositional inverse exists. Substitution of $q = \hat{V}(t)$ gives (1.2); conversely, the unique tangent-to-identity formal solution of (1.2) is that inverse.

For $n > 0$ let r_n be the unique positive solution of

$$r_n(1 + r_n)e^{2r_n} = an. \quad (2.4)$$

Define

$$j_n = \frac{n}{1 + r_n}, \quad k_n = \frac{nr_n}{1 + r_n}, \quad D_n = 2r_n^2 + 4r_n + 1, \quad S_n = \frac{\exp\left(\frac{nr_n(1 + 2r_n)}{1 + r_n}\right)}{\sqrt{D_n}}. \quad (2.5)$$

These are continuous saddle quantities; j_n need not be an integer. In particular, $r_n \sim \frac{1}{2} \log n$, $j_n \sim 2n/\log n$, and $k_n \sim n$.

Theorem 2.2 (Signed quadratic inverse universality). *Under (2.1),*

$$\boxed{v_n \sim -S_n e^{-br_n}}. \quad (2.6)$$

In particular, $v_n < 0$ for all sufficiently large n , including when some of the exceptional primitive weights are negative. For the pure model $w_j = 1$, $\lambda_j = aj^2$ for every j , this is precisely (1.1).

The theorem concerns formal coefficients. For $t > 0$ and $q \neq 0$, the quadratic tail in (1.2) does not even tend to zero in modulus. Thus no claim about an ordinary positive-real inverse obtained by summing that kernel is implicit in the theorem.

Editorial note (ProveIt, 2026-09-29). A second, independent claimed proof of (1.1) is [6], Theorem `thm:main`, written from the same snapshot without knowledge of this article (neither cites the other; neither has been independently reviewed). It inverts the kernel first, groups the exact tree expansion by large actions and merges multiple large actions in a positive majorant. Its hypotheses differ and neither set contains the other: it requires nonnegative changes of finitely many weights and slopes, but allows eventual slopes $aj^2 + dj + e$, with factor $\exp((d - \lambda_1 - w_2)r_n/a)$, which is (2.6) at $d = 0$. Its exact inverse coefficients equal the ones recorded here at every common degree (through 220 for $a = 1$ and 180 for $a = 2$), and its positive-ray Borel equivalent is the one in Theorem 2.4. It also proves the sharp relative error of every fixed core with at least two actions and the constant $e^{\kappa/(2a)}$ for a slope perturbation $\kappa j / \log j$; the signed exceptions, the one-action criterion of Theorem 2.3 and the maximum-modulus equivalent are proved only here.

For a fixed $M \geq 1$, define the analytic core and its inverse by

$$H_M(q, t) = \sum_{j=1}^M w_j q^j e^{\lambda_j t}, \quad H_M(Q_M(t), t) = t, \quad A_M(t) = \frac{1}{\partial_q H_M(Q_M(t), t)}. \quad (2.7)$$

The local implicit function theorem applies because $\partial_q H_M(0, 0) = 1$. The formal one-tail response is

$$L_M(t) = A_M(t) \sum_{j>M} w_j Q_M(t)^j e^{\lambda_j t}, \quad \ell_{n,M} = [t^n] L_M(t). \quad (2.8)$$

This is a locally finite formal series, not a convergent sum at positive t .

Theorem 2.3 (The leading core is exactly two actions). *Under (2.1), every fixed $M \geq 2$ satisfies*

$$\ell_{n,M} \sim S_n e^{-br_n}, \quad v_n \sim -\ell_{n,M}. \quad (2.9)$$

For the one-action core,

$$\ell_{n,1} \sim S_n e^{-(\lambda_1/a)r_n}, \quad \frac{v_n}{-\ell_{n,1}} \sim e^{-(w_2/a)r_n}. \quad (2.10)$$

Hence $M = 2$ is the smallest sufficient core in the pure model. For a general finite perturbation, $M = 1$ is sufficient exactly when $w_2 = 0$.

The proof uses a larger *auxiliary* core to control the remainder. Theorem 2.3 is not obtained by assuming that a direct absolute bound works optimally at $M = 2$.

Theorem 2.4 (Sharp entire Borel growth). *The ordinary Borel transform*

$$B(z) = \mathcal{B}\hat{V}(z) := \sum_{n \geq 1} \frac{v_n}{n!} z^n \quad (2.11)$$

is entire. For $R > 0$ put

$$s = s(R) = \frac{\sqrt{1 + 4aR} - 1}{2}. \quad (2.12)$$

Then, as $R \rightarrow +\infty$,

$$\boxed{-B(R) \sim \max_{|z|=R} |B(z)| \sim \frac{\exp\left(\frac{se^{2s}}{a} - bs\right)}{\sqrt{2s+1}}.} \quad (2.13)$$

In particular,

$$\lim_{R \rightarrow \infty} \frac{\log \log \max_{|z|=R} |B(z)|}{\sqrt{R}} = 2\sqrt{a}, \quad (2.14)$$

and the ordinary positive-ray Borel–Laplace integral diverges for every positive Laplace parameter.

3 Exact formal inversion with a frozen parameter

3.1 A marked tail expansion

Fix $M \geq \max(M_0, 2)$ temporarily. Let $P_M(s, t)$ denote the analytic inverse of $q \mapsto H_M(q, t)$, so $H_M(P_M(s, t), t) = s$. The first variable s and the external parameter t must be kept separate. In particular, differentiation below is with respect to s , with t *frozen*.

Introduce a marker z for primitive actions larger than M and let $V_z(t)$ solve

$$H_M(V_z(t), t) + z \sum_{j>M} V_z(t)^j e^{aj^2 t} = t. \quad (3.1)$$

For ordered tuples $\mathbf{h} = (h_1, \dots, h_m)$ with $h_i > M$, write

$$J = \sum_i h_i, \quad \Lambda = a \sum_i h_i^2, \quad \mathcal{D}_{m,J}(t) = \frac{1}{m!} \partial_s^{m-1} \left(P_{M,s}(s, t) P_M(s, t)^J \right) \Big|_{s=t}. \quad (3.2)$$

Lemma 3.1 (Exact core-resummed expansion). *Coefficientwise in t ,*

$$V_z(t) = Q_M(t) + \sum_{m \geq 1} (-z)^m \sum_{h_1, \dots, h_m > M} e^{\Lambda t} \mathcal{D}_{m,J}(t). \quad (3.3)$$

The contribution with $m = 1$ is $-zL_M(t)$. Moreover, $\mathcal{D}_{m,J}(t)$ is divisible by t^{J-m+1} .

Proof. Put $s = H_M(V_z(t), t)$ in (3.1). Then $s = t - zT(P_M(s, t), t)$, with $T(q, t) = \sum_{j>M} q^j e^{aj^2 t}$. Lagrange's expansion around $s = t$, applied to $P_M(s, t)$, gives

$$P_M(s, t) = P_M(t, t) + \sum_{m \geq 1} \frac{(-z)^m}{m!} \partial_s^{m-1} (P_{M,s}(s, t) T(P_M(s, t), t)^m) \Big|_{s=t}.$$

Expanding the m th power gives (3.3). For a completely formal justification, truncate the primitive action index and the desired t -degree first, perform the finite formal calculation, and then remove the truncation. Since $P_M(s, t)$ is divisible by s , the differentiated product is divisible by s^{J-m+1} . After $s = t$, a tuple can contribute to $[t^n]$ only if $J - m + 1 \leq n$. With $h_i \geq M + 1$, this implies $mM + 1 \leq n$ and bounds every h_i . Thus all relevant sums are finite. Finally $P_{M,s}(t, t) = A_M(t)$, proving the $m = 1$ assertion. $\square$

This formula is where the inverse cancellation is handled. The coefficients of P_M , and therefore of $\mathcal{D}_{m,J}$, may have either sign. We do not replace the core by the absolute values of its Taylor coefficients. We estimate its analytic values on a shrinking circle only after it has been inverted exactly.

3.2 An independent finite identity in the pure model

The pure model admits a useful separate check. Set $q = e^{-at}z$ in (1.2). Then

$$t = z + \sum_{d \geq 1} z^{d+1} e^{ad(d+1)t}. \quad (3.4)$$

Inverting in z by Lagrange, while keeping the t in the exponentials fixed, yields

$$\begin{aligned} v_n &= \frac{(-a)^{n-1}}{(n-1)!} \\ &+ \sum_{D=1}^{n-1} \sum_{m \geq 1, d_1 + \dots + d_m = D} \frac{(-1)^m (D+m)!}{m! (D+1)!} \frac{[a(\sum_i d_i (d_i + 1) - 1)]^{n-D-1}}{(n-D-1)!}. \end{aligned} \quad (3.5)$$

Indeed the m th perturbation contributes $(-1)^m \partial_z^{m-1}(z^{D+m})/m!$ at $z = t$, followed by multiplication by e^{-at} . Formula (3.5) is finite and exact, but it is not a stable numerical large-order algorithm: substantial signed cancellation remains. Its role in the package is independent low-order verification of the direct triangular recurrence.

4 The bare action saddle

For integers $1 \leq j \leq n$, put $k = n - j$ and

$$T_{n,j} = \frac{(aj^2)^k}{k!}, \quad Z_n = \sum_{j=1}^n T_{n,j}. \quad (4.1)$$

At $k = 0$ the expression is 1. These positive quantities are comparison weights, not inverse coefficients.

Lemma 4.1 (Bare saddle and slowly varying tilt). *For every fixed real β ,*

$$Z_n \sim S_n, \quad \sum_{j=1}^n T_{n,j} \exp\left(-\beta \frac{n-j}{j}\right) \sim S_n e^{-\beta r_n}. \quad (4.2)$$

The relevant action window is centered at j_n , with variance

$$\sigma_n^2 = \frac{k_n}{D_n} \sim \frac{2n}{(\log n)^2}. \quad (4.3)$$

For any fixed $C > 0$, comparison weights multiplied by $\sqrt{n+1} \exp(Cj + Cn/j)$ have exponentially negligible total mass outside

$$\mathcal{I}_n = \left[\frac{n}{2 \log n}, \frac{8n}{\log n} \right] \quad (4.4)$$

when their coefficient radii $k/(aj^2)$ are bounded by a sufficiently small fixed constant. Indices for which those radii are larger can instead be bounded at a fixed radius by $\exp(O(n))$.

Proof. Stirling's formula, away from $k = 0$, gives

$$T_{n,j} = \frac{e^{f_n(j)}}{\sqrt{2\pi(n-j)}} \left(1 + O((n-j)^{-1})\right), \quad f_n(j) = (n-j) \left(1 + \log \frac{aj^2}{n-j}\right). \quad (4.5)$$

Differentiation gives

$$f'_n(j) = -\log \frac{aj^2}{n-j} + \frac{2(n-j)}{j}, \quad (4.6)$$

$$f''_n(j) = -\frac{4}{j} - \frac{2(n-j)}{j^2} - \frac{1}{n-j} < 0. \quad (4.7)$$

Thus there is a unique stationary point. Setting $r = (n-j)/j$ in $f'_n(j) = 0$ gives (2.4), hence $j = j_n$. At that point,

$$f_n(j_n) = \frac{nr_n(1+2r_n)}{1+r_n}, \quad -f''_n(j_n) = \frac{D_n}{k_n}. \quad (4.8)$$

On $j \asymp n/\log n$, the third derivative is $O(n/j^3)$ and the fourth is $O(n/j^4)$. Therefore, for $j = j_n + \sigma_n x$ with $|x| \leq \log n$,

$$f_n(j) = f_n(j_n) - x^2/2 + O((1+|x|)^3/\sqrt{n}).$$

The Stirling prefactor is uniform there. Summation of the lattice Gaussian, whose mesh is $1/\sigma_n \rightarrow 0$, gives

$$\sum_j T_{n,j} \sim \frac{e^{f_n(j_n)}}{\sqrt{2\pi k_n}} \sqrt{2\pi} \sigma_n = S_n.$$

Strict concavity and (4.7) bound the portion of $\mathcal{I}_n$ outside that window by a polynomial factor times $e^{-c(\log n)^2} S_n$.

For the tilt, put $r(j) = n/j - 1$. In the same window,

$$|r(j) - r_n| \leq C \frac{n}{j_n^2} \sigma_n \log n = O((\log n)^2 / \sqrt{n}) = o(1).$$

Thus the tilt is uniformly $e^{-\beta r_n}(1 + o(1))$ in the central window. On $\mathcal{I}_n$ it is at most a fixed power of n , so the Gaussian tail bound remains sufficient.

Here are details of the coarser localization, also needed below. Write $x = j \log n / n$. Uniformly for x in a fixed compact subinterval of $(0, \infty)$,

$$\frac{f_n(j)}{n} = \log n - 2 \log \log n + \log a + 1 + 2 \log x - x + o(1). \quad (4.9)$$

The function $\phi(x) = 2 \log x - x$ has its unique maximum at $x = 2$ and a strictly smaller value on $x \leq 1/2$ or $x \geq 8$. To make the off-window comparison uniform, put $\theta = j/n$ and $L = \log n$. There is the exact identity

$$\begin{aligned} \frac{f_n(j)}{n} &= L - 2 \log L + \log a + 1 + \phi(x) + E_a(\theta), \\ E_a(\theta) &= -2\theta \log \theta - \theta(\log a + 1) - (1 - \theta) \log(1 - \theta). \end{aligned} \quad (4.10)$$

With $0 \log 0 = 0$, E_a is bounded on $[0, 1]$ and tends to zero as $\theta \downarrow 0$. In the small-radius range $k/(aj^2) \leq \delta$, either $j \geq n/2$ or $j \geq \sqrt{n/(2a\delta)}$. Thus the decoration, divided by n , is $C\theta + C/j + o(1)$, with $C/j = o(1)$ uniformly. For $x \leq 1/2$ and for $8 \leq x \leq K$, where K is fixed, both $E_a(\theta)$ and $C\theta$ tend uniformly to zero. The strict loss of ϕ proves the desired bound there. Choose $K > 8$ so large that $\phi(K) + \sup E_a + C < \phi(2) - 1$. As ϕ is decreasing after 2, (4.10) gives a loss at least $n/2$ for $x \geq K$, for large n . Together these estimates give a loss at least ηn outside (4.4). The Stirling prefactor costs only a polynomial; the endpoint $k = 0$ is handled directly and contributes an exponential decoration, still negligible compared with S_n .

Finally, if $k/(aj^2) > \delta$ for fixed $\delta > 0$, then $j = O(\sqrt{n})$. A Cauchy bound at radius δ has logarithm at most $n \log(1/\delta) + aj^2 \delta + O(j) = O(n)$. This is negligible compared with $\log S_n = n \log n - 2n \log \log n + O(n)$. This proves both the localization statements and (4.2). $\square$

The only use of the last, decorated assertion will have a fixed C . No uniformity as the core or the model parameters vary with n is being asserted.

5 A signed finite-core reduction with arbitrary algebraic accuracy

5.1 Core bounds independent of the number of retained actions

The first jet of $P_M(s, t)/s$ is independent of M once $M \geq 2$:

$$\log \frac{P_M(s, t)}{s} = -\lambda_1 t - w_2 s + O((|s| + |t|)^2). \quad (5.1)$$

For each fixed M there is consequently a radius $\delta_M > 0$ on which

$$|P_M(s, t)| \leq |s| \exp(C_0(|s| + |t|)), \quad |P_{M,s}(s, t)| \leq 2. \quad (5.2)$$

The radius may depend on all the retained data, but the displayed constant $C_0 = 1 + |\lambda_1| + |w_2|$ does not grow with M . This distinction permits us to choose a larger auxiliary core without a circular dependence of the power-saving exponent on that choice.

Lemma 5.1 (Derivative estimate). *For every fixed sufficiently large M , every tuple in (3.2), and every sufficiently small $\tau > 0$,*

$$\sup_{|t|=\tau} |\mathcal{D}_{m,J}(t)| \leq \frac{4^m J^{m-1}}{m!} \tau^{J-m+1} \exp(3C_0 J \tau). \quad (5.3)$$

The smallness threshold for τ depends on M but not on the tuple.

Proof. For $m = 1$, this follows directly from (5.2). For $m \geq 2$, take a Cauchy circle in the s variable centered at $s = t$ with radius $\rho = (m-1)\tau/J$. Since $J \geq m(M+1)$, this circle lies in $|s| \leq 2\tau$. On it,

$$|P_M(s, t)|^J \leq \tau^J \left(1 + \frac{m-1}{J}\right)^J e^{3C_0 J \tau} \leq \tau^J e^{m-1+3C_0 J \tau}.$$

Cauchy's derivative estimate and the factor $1/m!$ in (3.2) give

$$|\mathcal{D}_{m,J}| \leq \frac{2}{m} \left(\frac{eJ}{m-1}\right)^{m-1} \tau^{J-m+1} e^{3C_0 J \tau}.$$

Since $(m-1)! \leq (m-1)^{m-1}$ and $2e^{m-1} \leq 4^m$, this is (5.3). $\square$

5.2 Merging the tail actions

For a tuple let

$$d_i = h_i - 1 \geq M, \quad D = \sum_i d_i, \quad j = D + 1 = J - m + 1, \quad k = n - j. \quad (5.4)$$

The effective single action j has the same minimal t -degree as the tuple. Its quadratic excess over the tuple is

$$\Delta = j^2 - \sum_i h_i^2 = D^2 - \sum_i d_i^2 - (m-1) = 2 \sum_{i < l} d_i d_l - (m-1) \geq 0. \quad (5.5)$$

This exact identity is the tail-suppression mechanism.

Lemma 5.2 (Coefficient bound by a merged action). *For $j \in \mathcal{I}_n$, $m \geq 1$, and all sufficiently large n ,*

$$\left| [t^n] e^{\Lambda t} \mathcal{D}_{m,J}(t) \right| \leq C \sqrt{n+1} T_{n,j} \frac{4(8j)^{m-1}}{m!} \exp(6C_0 j \tau) \exp(-a\tau \Delta), \quad \tau = \frac{k}{aj^2}. \quad (5.6)$$

Here and below constants can depend on the fixed model and core.

Proof. In (5.3), $J = j + m - 1 \leq 2j$. Cauchy's coefficient bound at $|t| = \tau$ is therefore at most

$$\frac{4(8j)^{m-1}}{m!} \tau^{-k} e^{a(j^2 - \Delta)\tau + 6C_0 j \tau}.$$

At $\tau = k/(aj^2)$,

$$\frac{\tau^{-k} e^k}{T_{n,j}} = \frac{k! e^k}{k^k} \leq C \sqrt{k+1}$$

by Stirling's bound. On $\mathcal{I}_n$, $\tau \rightarrow 0$, so the analytic estimates apply uniformly for large n . $\square$

5.3 One giant tail action versus several

We now sum the bounds without assuming signs of any core coefficients. Let $L = \max_i d_i$ and $E = D - L$. If $L \geq D/2$ and $m \geq 2$, then $E \geq M$ and

$$\Delta \geq 2LE - (m - 1) \geq (D - 1/M)E \geq jE/2 \quad (5.7)$$

for large j . On $\mathcal{I}_n$, $k/j \geq (\log n)/9$ eventually. Hence

$$e^{-a\tau\Delta} \leq n^{-E/20}. \quad (5.8)$$

The constants 9 and 20 are deliberately loose.

For fixed m , choose a position for the largest component (at most m choices) and sum the other $m - 1$ components. Dropping the upper bound on their total only enlarges the bound. Put

$$z_n = \sum_{d \geq M} n^{-d/20} = \frac{n^{-M/20}}{1 - n^{-1/20}} \leq 2n^{-M/20}$$

for sufficiently large n . The sum of the tuple factors in (5.6) with a giant and $m \geq 2$ is at most

$$4 \sum_{m \geq 2} \frac{(8jz_n)^{m-1}}{(m-1)!} = 4(e^{8jz_n} - 1) = O(jn^{-M/20}) \quad (5.9)$$

when $M > 20$.

If no component is at least $D/2$, then $\sum_i d_i^2 \leq LD < D^2/2$ and $m - 1 \leq D/M$. Thus $\Delta \geq j^2/3$ for large j . The suppression in (5.6) is at most $e^{-k/3}$. There are at most 2^D ordered compositions of D , and

$$\sum_{m \geq 1} \frac{4(8j)^{m-1}}{m!} \leq 4e^{8j}.$$

The total tuple factor in the no-giant case is consequently at most $4e^{9j-k/3} = e^{-\eta n}$ on $\mathcal{I}_n$ for large n . All these estimates allow overcounting ties for the largest component.

Finally, on $\mathcal{I}_n$,

$$e^{6C_0j\tau} = \exp\left(\frac{6C_0}{a} \frac{k}{j}\right) \leq n^{12C_0/a}. \quad (5.10)$$

Combining (5.6)–(5.10), summing over j , and using Lemma 4.1, bounds the multi-tail contribution in the coarse window by

$$CS_n n^{3/2+12C_0/a-M/20} + S_n e^{-\eta n}. \quad (5.11)$$

5.4 Off-window contributions and the finite-core theorem

Outside $\mathcal{I}_n$, first ignore the nonnegative merger defect. The sum of all tuple combinatorial factors with fixed j is bounded by e^{Cj} , by the same composition estimate just used. In the small-radius range the total coefficient majorant is therefore

$$C\sqrt{n+1} T_{n,j} \exp(Cj + Cn/j).$$

Lemma 4.1 makes its off-window sum exponentially negligible relative to S_n . In the complementary range $j = O(\sqrt{n})$, use a fixed Cauchy radius small enough for the core instead. The logarithm of the total bound is $O(n)$, again negligible. The coefficient of the analytic series Q_M is also at most exponential in n .

Theorem 5.3 (Arbitrary algebraic accuracy after a finite core). *For every fixed $A > 0$ there is a fixed M such that*

$$v_n + \ell_{n,M} = O\left(S_n n^{-A-|b|}\right). \quad (5.12)$$

One sufficient, non-optimal choice is any integer $M \geq \max(M_0, 2)$ satisfying

$$\frac{M}{20} > A + 3 + \frac{12C_0}{a} + |b|. \quad (5.13)$$

Proof. Use the exact expansion (3.3) at $z = 1$. The $m = 1$ term is $-L_M$. The remaining tuple terms satisfy (5.11) in $\mathcal{I}_n$ and the exponentially smaller estimates outside it. The strict inequality in (5.13) absorbs the factor $n^{3/2}$ and leaves more than the required saving. The analytic core coefficient is negligible on the same scale. $\square$

Remark 5.4 (Why this proves a signed statement). The forward coefficients never enter the proof. The inverse response itself has size S_n times a possibly small power of n ; a merely $o(S_n)$ error would not suffice. The explicit saving in (5.12) beats that polynomial suppression. This is exactly the obstacle avoided by using a sufficiently large auxiliary core before reducing to the minimal core.

6 Evaluation of the one-tail inverse response

For every fixed $M \geq 2$, the diagonal core has expansion

$$Q_M(t) = t - ct^2 + O(t^3), \quad A_M(t) = 1 + O(t). \quad (6.1)$$

Consequently, in a fixed disk about zero,

$$\log \frac{Q_M(t)}{t} = -ct + R_M(t), \quad R_M(t) = O(t^2). \quad (6.2)$$

For large tail actions the one-tail summand is

$$A_M(t)Q_M(t)^j e^{aj^2 t} = t^j e^{\alpha_j t} E_j(t), \quad \alpha_j = aj^2 - cj, \quad E_j(t) = A_M(t)e^{jR_M(t)}. \quad (6.3)$$

For all sufficiently large j , $\alpha_j > 0$, even when c is negative.

Lemma 6.1 (Uniform inner coefficient estimate). *Uniformly for integers $j \in \mathcal{I}_n$, with $k = n - j$,*

$$[t^k]e^{\alpha_j t} E_j(t) = \frac{\alpha_j^k}{k!} \left(1 + O\left(\frac{(\log n)^3}{n}\right) \right). \quad (6.4)$$

Proof. Use the radius $\rho = k/\alpha_j$. On $\mathcal{I}_n$, $\rho = O((\log n)^2/n)$ and $j\rho^2 = O((\log n)^3/n)$. By analyticity and (6.2),

$$\sup_{|t|=\rho} |E_j(t) - 1| \leq C(\rho + j\rho^2) =: \varepsilon_n.$$

The Cauchy integral for the error is at most

$$\varepsilon_n \rho^{-k} \frac{1}{2\pi} \int_{-\pi}^{\pi} e^{k \cos \theta} d\theta \leq C \varepsilon_n \rho^{-k} \frac{e^k}{\sqrt{k+1}}.$$

The last bound follows from $1 - \cos \theta \geq 2\theta^2/\pi^2$ on $[-\pi, \pi]$. Stirling's lower bound shows that this is at most $C\varepsilon_n \alpha_j^k/k!$. The coefficient with $E_j = 1$ is exactly $\alpha_j^k/k!$. Notice that the entire factor e^{-cjt} was absorbed into α_j before taking absolute values; failing to do this would lose the polynomial cancellation we need. $\square$

Since $\alpha_j/(aj^2) = 1 - b/j$, on $\mathcal{I}_n$ we have

$$\frac{\alpha_j^k}{(aj^2)^k} = \exp\left(-b\frac{k}{j} + O(k/j^2)\right) = e^{-bk/j} \left(1 + O((\log n)^2/n)\right). \quad (6.5)$$

Summing (6.4) and using Lemma 4.1 gives

$$\ell_{n,M} \sim S_n e^{-br_n}. \quad (6.6)$$

The off-window terms are controlled by the $m = 1$ version of the analytic majorant used in Section 5. Finitely many nonquadratic or exceptional summands are analytic at zero and contribute only exponentially growing coefficients. Thus (6.6) holds for *every* fixed $M \geq 2$, even if that core does not contain all exceptional actions.

Proof of Theorems 2.2 and 2.3. Choose the auxiliary M from Theorem 5.3, for instance with $A = 1$. Since $r_n < \log n$ eventually, $e^{-br_n} \geq n^{-|b|}$. Hence the error in (5.12) is $o(S_n e^{-br_n})$. Combine it with (6.6) to obtain (2.6). The same one-tail equivalent for any other fixed $M \geq 2$ then proves (2.9) without a second remainder estimate at that smaller core.

For $M = 1$, $Q_1(t) = te^{-\lambda_1 t}$ and $A_1(t) = e^{-\lambda_1 t}$. The preceding one-tail calculation applies with $c = \lambda_1$; the additional factor $A_1(t) = 1 + O(t)$ does not alter the equivalent. Division by (2.6) gives (2.10). As $r_n \rightarrow \infty$, its right side tends to 1 exactly for $w_2 = 0$, to 0 for $w_2 > 0$, and to $+\infty$ for $w_2 < 0$. $\square$

6.1 What the suppression factor measures

At the bare inner saddle $t \approx k/(aj^2)$,

$$j \log(Q_M(t)/t) = -cjt + O(jt^2) = -\frac{c}{a} \frac{k}{j} + o(1).$$

The outer saddle has $k/j = r_n$. This gives the entire leading penalty e^{-br_n} . Only the quadratic coefficient of the inverse core survives. All higher diagonal core coefficients are multiplied by quantities beginning with $jt^2 = O((\log n)^3/n)$.

This mechanism differs from simply substituting the forward coefficient asymptotic into a few inverse-composition terms. In the pure model, the repository's forward result is $u_n \sim S_n \exp((1 + 1/a)r_n^2)$ [1]. *Combining that external forward theorem with our independent inverse theorem* gives

$$\frac{-v_n}{u_n} \sim \exp(-(1 + 1/a)(r_n^2 + r_n)). \quad (6.7)$$

The forward theorem is needed only for this comparison, not for any of our inverse or Borel proofs.

6.2 An explicit two-action formula

For the pure model,

$$Q_2(t) = \frac{e^{-3at}}{2} \left(\sqrt{1 + 4te^{2at}} - 1 \right), \quad A_2(t) = \frac{e^{-at}}{\sqrt{1 + 4te^{2at}}}. \quad (6.8)$$

These are the branches analytic at zero. Let $C(z) = (1 - \sqrt{1 - 4z})/(2z)$ be the Catalan series. Then $Q_2 = te^{-at}C(-te^{2at})$, and the elementary identity

$$\frac{C(z)^j}{\sqrt{1 - 4z}} = \sum_{h \geq 0} \binom{2h + j}{h} z^h$$

(obtainable by Lagrange inversion) gives the finite coefficient formula

$$\ell_{n,2} = \sum_{j=3}^n \sum_{h=0}^{n-j} (-1)^h \binom{2h+j}{h} \frac{[a(j^2 - j - 1 + 2h)]^{n-j-h}}{(n-j-h)!}. \quad (6.9)$$

Thus a completely explicit finite sum already has the correct full inverse equivalent. It is still a signed sum; exact arithmetic or adequate working precision is necessary.

7 Consequences for coefficient growth and formal transseries

Corollary 7.1 (Ratio law and exact Gevrey threshold). *Under (2.1),*

$$\frac{v_n}{v_{n-1}} \sim e^{2r_n} \sim \frac{4an}{(\log n)^2}, \quad (7.1)$$

$$|v_n|^{1/n} \sim \frac{4an}{e(\log n)^2}, \quad (7.2)$$

$$\log \frac{|v_n|}{n!} = -2n \log \log n + n \log(4a) + o(n). \quad (7.3)$$

The inverse has radius of convergence zero. It is Gevrey-1 of zero exponential type: for every $A > 0$ there is $C_A < \infty$ with $|v_n| \leq C_A A^n n!$. It is not Gevrey- s for any $s < 1$.

Proof. Extend r_n smoothly to positive real n . Implicit differentiation gives

$$r'(n) = \frac{r(1+r)}{n(2r^2 + 4r + 1)}. \quad (7.4)$$

For $F(n) = nr(1+2r)/(1+r)$, the envelope identity is $F'(n) = 2r$. The logarithmic derivative of $\sqrt{2r^2 + 4r + 1}$ is $O(1/n)$ and $br'(n) = O(1/n)$. Thus the ratio of successive smooth equivalents is $e^{2r_n}(1+o(1))$. The two relative errors in Theorem 2.2 also have ratio tending to one. This proves (7.1).

Using (2.4),

$$\frac{1}{n} \log |v_n| = 2r_n - 1 + \frac{1}{1+r_n} + o(1), \quad e^{2r_n} = \frac{an}{r_n(1+r_n)}.$$

This proves (7.2); Stirling then gives (7.3). The conclusions follow directly. For $s < 1$, $\log(|v_n|/(n!)^s) = (1-s)n \log n - 2n \log \log n + O(n)$, which exceeds $n \log A$ eventually for every fixed A . $\square$

Corollary 7.2 (Flatness in every fixed positive factorial scale). *For all $\alpha > 0$, all real β , and every integer $K \geq 0$,*

$$v_n = o(\alpha^{n+\beta} \Gamma(n + \beta - K)). \quad (7.5)$$

Thus the gamma-scale asymptotic derivation of [3], with any fixed positive α , assigns the zero expansion to this divergent inverse.

Proof. The logarithm of the ratio in (7.5) equals $-2n \log \log n + O(n) + O(\log n)$, which tends to $-\infty$. This is exactly the all-zero case of the fixed gamma-scale definition. $\square$

Corollary 7.3 (Least formal term). *For $t > 0$, let $N(t)$ minimize $|v_n|t^n$ over sufficiently large indices past the eventual-sign threshold. Then, as $t \downarrow 0$,*

$$N(t) \sim \frac{(\log(1/t))^2}{4at}. \quad (7.6)$$

Proof. The terms eventually grow without bound for fixed t , by (7.2). Their successive ratios are asymptotic to te^{2r_n} , an increasing smooth comparison. At its crossing of 1, $r = \frac{1}{2} \log(1/t)$ and the corresponding real index is $r(1+r)/(at)$. The uniform eventual relative bounds furnished by (7.1) confine any minimizing index to a relative $o(1)$ window about this value. $\square$

This is a statement about formal terms, not a remainder estimate for an actual analytic realization. Substituting $t = e^{-x}$ gives an exponential transseries with a least-term index asymptotic to $x^2 e^x / (4a)$. No summability conclusion is inferred from this formal optimization.

8 Entire Borel growth: a second, explicit saddle

8.1 Passing from coefficient equivalents to an entire function

Equation (7.3) proves that B in (2.11) is entire. By Theorem 2.2, its coefficients are eventually negative. Let

$$\tilde{B}(R) = \sum_{n \geq 1} \frac{S_n e^{-br_n}}{n!} R^n.$$

All terms are positive, and their sum tends to infinity faster than every polynomial. The coefficient equivalent implies

$$-B(R) \sim \tilde{B}(R). \quad (8.1)$$

To see this without an unjustified interchange of limits, fix $\varepsilon > 0$. For every n beyond a fixed threshold, the ratio of the actual negative coefficient to the positive comparison coefficient is between $1 - \varepsilon$ and $1 + \varepsilon$. Sum those inequalities. The finitely many discarded terms form a polynomial, negligible relative to $\tilde{B}(R)$. Then let $\varepsilon \downarrow 0$.

8.2 Stationary point and determinant

For continuous $n > 0$, use $r = r(n)$ from (2.4) and define

$$h_R(n) = F(n) + n(1 - \log n) + n \log R, \quad F(n) = \frac{nr(1 + 2r)}{1 + r}. \quad (8.2)$$

By Stirling,

$$\frac{S_n e^{-br_n} R^n}{n!} = \frac{e^{h_R(n) - br_n}}{\sqrt{2\pi n D_n}} \left(1 + O(n^{-1})\right). \quad (8.3)$$

The exact derivative identities are

$$h'_R(n) = 2r - \log n + \log R = \log \frac{aR}{r(1+r)}, \quad (8.4)$$

$$h''_R(n) = -\frac{2r+1}{n(2r^2 + 4r + 1)} < 0. \quad (8.5)$$

Thus the saddle is characterized simply by $r = s$, where $s(1+s) = aR$. Its real index and exponent are

$$n_R = Re^{2s} = \frac{s(1+s)}{a} e^{2s}, \quad h_R(n_R) = \frac{se^{2s}}{a}. \quad (8.6)$$

The variance in the coefficient index is

$$\Sigma_R^2 = \frac{n_R(2s^2 + 4s + 1)}{2s + 1}. \quad (8.7)$$

Lemma 8.1 (The outer Laplace sum). *As $R \rightarrow \infty$,*

$$\tilde{B}(R) \sim \frac{e^{h_R(n_R)-bs}}{\sqrt{2s+1}}. \quad (8.8)$$

Proof. The width satisfies $\Sigma_R/n_R = O(\sqrt{s/n_R}) \rightarrow 0$. In a neighborhood of n_R with relative width tending to zero, (7.4) gives $r(n) - s = O(|n - n_R|/n_R)$. The amplitude $e^{-br}/\sqrt{2\pi nD}$ therefore has relative change $o(1)$ on a Gaussian window. Differentiating (8.5) gives $h_R'''(n) = O(1/(n_R^2 s))$ there, so $h_R''' \Sigma_R^3 = O(\sqrt{s/n_R}) \rightarrow 0$. Taylor expansion and lattice Gaussian summation yield

$$\tilde{B}(R) \sim \frac{e^{h_R(n_R)-bs}}{\sqrt{2\pi n_R(2s^2 + 4s + 1)}} \sqrt{2\pi} \Sigma_R = \frac{e^{h_R(n_R)-bs}}{\sqrt{2s+1}}.$$

For completeness, the tail localization follows from the global strict concavity (8.5). On $[n_R/2, 2n_R]$, $-h_R''$ is comparable to $1/(n_R s)$, giving Gaussian decay outside an expanding central window. At the endpoints the exponent loses at least a fixed multiple of n_R/s . For $n \geq 2n_R$, the decreasing derivative satisfies $h_R'(n) \leq h_R'(2n_R) \leq -c/s$ for large R ; thus the remaining sum is bounded by a geometric tail after its already negligible endpoint. The lower tail is handled by the analogous positive derivative to the left of $n_R/2$. The amplitudes grow or decay only by fixed powers of n and logarithms there, so they do not offset these exponential losses. This also justifies discarding finitely many small indices and using Stirling uniformly in the contributing range. $\square$

Proof of Theorem 2.4. Combine (8.1), Lemma 8.1, and (8.6). Because $v_n < 0$ eventually,

$$|B(R)| \leq \max_{|z|=R} |B(z)| \leq \sum_{n \geq 1} |v_n| R^n / n! = -B(R) + P(R),$$

where P is a fixed polynomial accounting for the finitely many coefficients with the other sign. Since $-B(R)$ has the growth already proved, $P(R) = o(-B(R))$, which gives the maximum-modulus equivalent.

Taking logarithms twice in (2.13), the leading expression is $2s + \log(s/a) + o(1)$. As $s \sim \sqrt{aR}$, this proves (2.14). In our ordinary Borel convention the positive-ray Laplace expression would be

$$\frac{1}{t} \int_0^\infty e^{-R/t} B(R) dR, \quad t > 0.$$

The integrand is eventually negative, and $\log(-B(R)) - R/t \rightarrow +\infty$. It therefore does not tend to zero and its integral diverges. The same obstruction rules out positive-direction ordinary 1-summability; analyticity alone is not enough without the required exponential-growth bound [4]. $\square$

8.3 Interpretation and limits

The Borel transform has no finite singularities at all. Nevertheless it cannot be integrated by the ordinary positive-direction Laplace kernel. The obstruction is its growth at infinity. This supplies a concrete example where exact Gevrey-1 regularity, an entire Borel transform, and failure of ordinary directional Borel summation coexist.

The formula does not classify other complex directions. Cancellations along nonreal rays can substantially change growth. Nor does it identify a unique acceleration procedure or a canonical sectorial realization. Those are separate analytic problems, listed below.

9 Exact computations and diagnostic evidence

9.1 Independent algebraic routes

The package computes the inverse through degree 220 in four fixed models, using integer exponential normalization $V_n = n!v_n$. For integer weights and slopes these are integers. Let $P_{j,n} = n![t^n]\widehat{V}(t)^j$. The identities

$$P_{j,n} = \sum_{k=1}^{n-j+1} \binom{n}{k} V_k P_{j-1,n-k} \quad (j \geq 2), \quad (9.1)$$

$$V_n = - \sum_{j=1}^n w_j \sum_{k=j}^{n-1} \binom{n}{k} \lambda_j^{n-k} P_{j,k} \quad (n \geq 2), \quad (9.2)$$

with $V_1 = P_{1,1} = 1$, form a triangular exact algorithm. Formula (9.2) is just the coefficient of (1.2), with the unique unknown V_n isolated.

The first independent check uses (3.5) through degree 12 for $a = 1$ and $a = 2$ (24 assertions). The second constructs the forward coefficients independently by the positive Lagrange identity

$$u_n = \sum_{\substack{j_1+\dots+j_m=n \\ j_i \geq 1, m \geq 1}} \frac{(a \sum_i j_i^2)^{m-1}}{m!} \quad (9.3)$$

and checks $\widehat{U}(\widehat{V}(t)) = t$ through degree 11 in both models (two block assertions). A separate symbolic script verifies six identities: the action curvature, the coefficient envelope derivative, the Borel curvature and stationary value, and two core-jet formulas. All 32 recorded assertions pass. These are finite exact checks, not formal proofs of the limit theorems.

9.2 What the numerical ratios show

Table 1 uses exact coefficients followed by 80-digit evaluation of the smooth comparison. The errors shown are not interval certificates. The two-core response is evaluated independently using (6.9).

Table 1: Pure quadratic feedback. The predicted limit in both ratio columns is 1.

a	n	r_n	$-v_n/(S_n e^{-(1+1/a)r_n})$	$-v_n/\ell_{n,2}$
1	40	1.298005	0.607745	0.855019
1	80	1.519691	0.725965	0.908128
1	120	1.654076	0.780419	0.928350
1	160	1.751348	0.813338	0.939697
1	220	1.860772	0.844814	0.950076
2	40	1.519691	0.656706	0.930063
2	80	1.751348	0.768248	0.961094
2	120	1.890976	0.817596	0.971321
2	160	1.991732	0.846760	0.976658
2	220	2.104802	0.874148	0.981282

The main equivalent has substantial finite- n corrections even at degree 220; the table must not be described as near-perfect asymptotic agreement. The exact two-core expression absorbs more of those corrections and is noticeably closer. Neither observation replaces the proof of tail dominance.

Two additional tests change only finitely many primitive data. In the first, $a = 1$, $w_2 = -2$, $w_3 = 3$, $\lambda_1 = 0$, and $\lambda_2 = 7$, with the other data pure. Here $c = -2$: the leading core factor is an enhancement e^{2r_n} rather than a suppression. The ratio to the predicted equivalent is about 2.693 at $n = 220$, so the asymptotic regime is visibly slow. In the second, $a = 1$, $w_2 = -1$ and all other data are pure, so $c = 0$ and the leading penalty cancels. The exact data for both models are included in `data/diagnostics.csv`; they are stress tests of the signed-core extension rather than evidence for a uniform onset bound.

9.3 Reproduction and numerical safeguards

Run

```
python code/verify.py --order 220
python code/symbolic_checks.py
pdflatex -interaction=nonstopmode -halt-on-error article.tex
pdflatex -interaction=nonstopmode -halt-on-error article.tex
```

The exact coefficient algorithm needs only Python’s standard library. `mpmath` is used for the comparison values, and `sympy` for the six symbolic checks. The scripts validate the model normalization and retain all coefficient signs. The ordinary coefficients are recovered by dividing stored integers by $n!$; confusing the two normalizations would invalidate both the ratios and the Borel transform.

The full exact algorithm is cubic in the truncation order in arithmetic operation count and uses a quadratic table. Its bit cost is larger because coefficient sizes increase rapidly. No complexity claim suppresses that integer-arithmetic cost.

10 Further research questions and topics

Research question 10.1 (Effective onset and small- a uniformity). Replace the asymptotic neighborhoods in the core estimates by computable radii and constants. The target is a certified pair of bounds for $-v_n/(S_n e^{-br_n})$ with an explicit onset depending on a and the exceptional data. The proof here is not uniform as $a \downarrow 0$; the factor $b = 1 + 1/a$ makes that limit especially delicate.

Research question 10.2 (An optimal signed remainder at the two-action core). Theorem 2.3 proves $v_n \sim -\ell_{n,2}$ but obtains it through a large auxiliary core. Determine the first term of $v_n + \ell_{n,2}$ and a sharp relative error. The natural next contribution is one additional primitive action 3, dressed by the two-action core; its cancellation and combinatorial factors should be kept exact.

Research question 10.3 (All-order inverse asymptotics). Derive explicit functions $P_k(r)$ for a full expansion $v_n \sim -S_n e^{-br_n} (1 + P_1(r_n)/n + P_2(r_n)/n^2 + \dots)$, with a precise meaning for the growing logarithmic coefficients. The finite-core theorem supplies arbitrary algebraic tail savings, but this paper does not perform the complete two-saddle expansion or claim the polynomial form of every P_k without proof.

Research question 10.4 (Which primitive data first enter each correction?). The leading term depends only on $\lambda_1 + w_2$. The next logarithmic core jet is $2\lambda_1 w_2 + \frac{3}{2}w_2^2 - w_2 \lambda_2 - w_3$. Determine the exact filtration of primitive data in successive inverse asymptotic coefficients. Such a theorem would turn finite-core universality into an accuracy-dependent information budget.

Research question 10.5 (Subquadratic feedback). Extend the signed reduction to $\lambda_j = aj^p$ with $1 < p < 2$. The quantity jt at the inverse saddle then grows faster than logarithmically, so a

polynomial saving from a fixed crude majorant need not beat the core cancellation. The quadratic proof identifies the obstruction rather than solving it. A contour estimate that preserves more of the inverse core's phase may be necessary.

Research question 10.6 (Regularly varying perturbations of the quadratic tail). For $\lambda_j = aj^2L(j)$ with a controlled slowly varying function L , determine the correct merger-defect estimate and inverse penalty. The exact identity (5.5) is no longer available, but quantitative convexity and regular variation may replace it. Mere pointwise $\lambda_j/(aj^2) \rightarrow 1$ is not enough to guarantee a multiplicative equivalent.

Research question 10.7 (Complex-direction Borel growth). Determine an indicator-type description of $B(Re^{i\theta})$ on fixed and shrinking angular sectors. The maximum-modulus theorem follows from eventual one-sign coefficients, but does not identify the cancellation curves away from the positive axis. Locate directions where ordinary Laplace growth might hold and determine their boundaries.

Research question 10.8 (Acceleration at the Borel point at infinity). Construct a summation or acceleration procedure adapted to $\log(-B(R)) \sim se^{2s}/a$, $s \sim \sqrt{aR}$. A useful result would identify a canonical analytic realization on a specified sector and prove compatibility with formal inversion. The positive ordinary Borel obstruction alone neither chooses such a procedure nor proves uniqueness.

Research question 10.9 (A moving-scale asymptotic calculus). Build an algebraic asymptotic calculus for sequences whose ratio is $n/(\log n)^2$ times a constant, rather than a fixed factorial ratio. The vanishing of every fixed positive-gamma-scale invariant in (7.5) suggests the need for additional moving-scale data. Seek product and composition rules that recover the factor e^{-br_n} without a separate saddle calculation in each example.

Research question 10.10 (Negative-real analytic realization). For $t < 0$, the Gaussian decay of e^{aj^2t} makes the primitive tail convergent for fixed q . Analyze the small solution of (1.2) in complex neighborhoods of the negative axis, prove an asymptotic expansion there, and compare it with any accelerated realization from the preceding question. Domain control near $t = 0$ must be included.

Research question 10.11 (Faster exact coefficient extraction). The triangular integer algorithm is cubic in arithmetic operations. Investigate fast truncated composition, divide-and-conquer recurrences, and parameter-dependent polynomial multiplication to reduce its cost. An effective implementation should report cancellation-sensitive working precision separately from algebraic operation counts.

Research question 10.12 (A staged Lean formalization). Begin with local finiteness, the frozen-parameter Lagrange identity, the merger defect, and the ordered-composition majorant. Then formalize analytic core estimates and the two saddle lemmas. These stages separate finite algebra, inequalities, and complex analysis; none is represented here as already checked by a proof assistant.

11 Conclusion and proof boundary

The signed quadratic inverse conjecture can be resolved without pretending that its coefficients are positive. The method first sums the background cancellation into an analytic inverse core, then pays for all remaining large-action configurations with the exact quadratic merger defect. A large auxiliary core makes that payment arbitrarily small on the relevant polynomial scale. A separate one-tail saddle calculation then reduces the leading answer to just two primitive actions.

The resulting equivalent is stable under arbitrary real finite perturbations, with its leading exponential correction determined by $\lambda_1 + w_2$. It implies eventual negativity, a precise logarithmically damped factorial growth scale, and an explicit entire Borel maximum-modulus equivalent. The Borel obstruction occurs at infinity. None of these conclusions supplies a general acceleration theorem, an optimal finite- n certificate, a uniform subquadratic theory, or a completed Lean development.

A Core jets and the source of finite-order corrections

For $M \geq 3$, expand $Q_M(t) = t - ct^2 + q_3t^3 + O(t^4)$. Substitution into $H_M(Q_M(t), t) = t$ gives

$$c = \lambda_1 + w_2, \tag{A.1}$$

$$q_3 = \frac{\lambda_1^2}{2} + 3\lambda_1w_2 + 2w_2^2 - w_2\lambda_2 - w_3. \tag{A.2}$$

Hence

$$\log \frac{Q_M(t)}{t} = -ct + \left(2\lambda_1w_2 + \frac{3}{2}w_2^2 - w_2\lambda_2 - w_3\right)t^2 + O(t^3), \tag{A.3}$$

and

$$A_M(t) = 1 - (\lambda_1 + 2w_2)t + O(t^2). \tag{A.4}$$

In the pure model the quadratic logarithmic coefficient in (A.3) is $\frac{1}{2} - 2a$. At the inverse saddle its contribution has size $jt^2 = O((\log n)^3/n)$. This explains why the leading equivalent can converge slowly while the two-core formula performs better at moderate orders. These scale estimates are not a claimed complete first-correction formula: the action-saddle displacement and its determinant also contribute.

B Proof-dependency audit

Result	Essential inputs	Status
Formal inverse and marked response	Local finiteness; classical Lagrange inversion with frozen parameter	Proved here
Bare saddle	Stirling; strict concavity; lattice Gaussian sum	Proved here
Arbitrary-accuracy signed core reduction	Analytic core Cauchy estimate; exact merger defect; composition count; coarse localization	Proved here; constants not optimized
Inverse equivalent	Core reduction plus uniform one-tail saddle	Proved here
Minimal leading core	Same one-tail equivalent for every fixed $M \geq 2$; explicit $M = 1$ calculation	Proved here
Finite signed perturbations	Fixed analytic core; unchanged eventual quadratic tail	Proved here; not uniform in the perturbation data
Entire Borel equivalent	Coefficient theorem; eventual negativity; second real saddle	Proved here
Forward/inverse ratio (6.7)	Repository forward theorem plus present inverse theorem	Explicit external dependency

Result	Essential inputs	Status
Finite exact checks	Integer recurrence; independent composition formulas; symbolic identities	Executed; not asymptotic proofs
Lean verification, acceleration, nonreal Borel indicator	Additional work	Not supplied

The reference to a classical Lagrange formula is not a hidden appeal to a positivity theorem. All sign-sensitive analytic estimates are displayed in Sections 5 and 6. The two instances of asymptotic summation use positive comparison weights after the core cancellation has already been extracted. No numerical observation is an assumption in the proof chain.

C Notation guide

Symbol	Meaning
$\widehat{U}, \widehat{V}$	Formal forward series and its compositional inverse.
v_n, V_n	Ordinary inverse coefficient and integer normalization $V_n = n!v_n$.
a, w_j, λ_j	Positive quadratic-tail constant; real primitive weights and slopes.
c, b, C_0	$\lambda_1 + w_2$, c/a , and $1 + \lambda_1 + w_2 $.
H_M, P_M, Q_M, A_M	Finite kernel; its inverse in the first variable; diagonal inverse; inverse derivative.
$L_M, \ell_{n,M}$	Formal one-tail inverse response and its ordinary coefficient; the inverse contains $-L_M$.
m, h_i, J, Λ	Number of tail actions; tail indices; their sum; sum of their slopes.
d_i, D, j, k, Δ	Excess indices; total excess; merged action; coefficient complement; quadratic merger defect.
r_n, j_n, k_n, D_n, S_n	Bare saddle coordinate, action, complement, determinant factor, and coefficient equivalent.
B, R, s, n_R, Σ_R	Ordinary Borel transform; positive radius; explicit Borel saddle coordinate; index saddle; its width.

Here D is total excess, D_n a saddle determinant, and Δ the merger defect.

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